7 **Confidence intervals and tests using the *t* – distribution**	7.1	Hypothesis test and confidence interval for the mean of a Normal distribution with unknown variance.	Use of $\dfrac{\overline{X}-\mu}{\dfrac{S}{\sqrt{n}}} \sim t_{n-1}$ Candidates may use *t*-tables or their calculators to calculate critical or p-values.
	7.2	Paired *t*-test.	
	7.3	Hypothesis test and confidence interval for the difference between two means from independent Normal distributions when the variances are equal but unknown. Use of the pooled estimate of variance.	Use of *t*-distribution. $\dfrac{\overline{X}-\overline{Y}-(\mu_x-\mu_y)}{S\sqrt{\dfrac{1}{n_x}+\dfrac{1}{n_y}}} \sim t_{n_x+n_y-2}$, under H_0. $s^2 = \dfrac{(n_1-1)s_1{}^2+(n_2-1)s_2{}^2}{n_1+n_2-2}$